The Invariance Principle — problem 1.42

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The following game is played on an infinite chessboard. Initially, each cell of an $n \times n$ square is occupied by a chip. A move consists in a jump of a chip over a chip in a horizontal or vertical direction onto a free cell directly behind it. The chip jumped over is removed. Find all values of n, for which the game ends with one chip left over (IMO 1993 and AUO 1992 !).